

Supplementary Information

Temperature coefficients of perovskite/silicon tandem solar cells.

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Experimental procedures

Silicon heterojunction bottom cell fabrication

Float-zone Si wafers (Topsil, PV-FZ, 1-5 $\Omega\cdot\text{cm}$) with thicknesses of $280 \pm 20 \mu\text{m}$ are used for silicon heterojunction bottom cell fabrication. Pyramids on both sides with heights of less than $1\mu\text{m}$ are obtained with an alkaline solution. After texturing the wafers go through RCA cleaning and a final hydrofluoric acid bath before transferring to the plasma-enhanced chemical vapour deposition (PECVD, Indeotec Octopus II). Intrinsic, doped amorphous silicon, and nano-crystalline silicon are deposited at 200°C . 10nm of IZO is sputtered by physical vapour deposition (PVD) and serves as a recombination junction on top of the nc-Si:H(n). 150nm of ITO and 250nm of Ag are sputtered on the rear side.

Perovskite top cell fabrication

The silicon bottom cells are subjected to a 10-minute UV-Ozone treatment before being transferred to an N₂-filled glovebox. 2PACz (1 mg/ml, ethanol) is spin-coated at 5000 rpm for 30 s, followed by drying at 100°C for 8 minutes. The triple cation perovskite solution Cs_{0.05}FA_{0.8}MA_{0.15}Pb(I_{0.75}Br_{0.25})₃ (1.7 M, DMF:DMSO 4:1 v/v) is prepared by dissolving a mixture of FAI, MABr (Greatcell Solar Materials), CsI (Sigma Aldrich), PbI₂, and PbBr₂ (Alfa Aesar). The bandgap of the perovskite is 1.68 eV. This perovskite precursor solution is spin-coated on the bottom cells using a two-step spin-coating process: initially spinning at 2,000 rpm and then at 7,000 rpm. During the acceleration between the two steps, chlorobenzene is deposited as a solvent quencher. The films are then annealed in the glovebox at 100 °C for 15 min. 20 nm of C₆₀ (nano-C) is then thermally evaporated (Angstrom EvoVac) on top of the perovskite. A 20-nm-thick layer of tin oxide was deposited by atomic layer deposition, using tetrakis(dimethylamino)tin and H₂O as the precursors with N₂ as the gas carrier. As the top electrode, 70 nm of IZO is sputtered in an Angstrom EvoVac sputtering system (base pressure <1 × 10⁻⁶ Torr). To functionally contact the transparent electrode, 350 nm of silver contacts were thermally evaporated (Angstrom EvoVac) (base pressure 1 × 10⁻⁶ Torr). Finally, a 100 nm MgF₂ antireflective coating was deposited onto the finished device by thermal evaporation. The device area of the silicon-perovskite tandem is 1 cm².

Solar cell characterization and outdoor testing

J-V measurements are performed under an LED-based solar simulator (Wavelabs Sinus 220). The intensities of the blue LED channels are modified to obtain different spectra. The resulting spectral irradiances are measured by the Wavelabs internal spectrophotometer. The unencapsulated devices are measured using an aperture area and our homemade program controlling a Keithley 2400 series SourceMeter. The device temperature is controlled using a vacuum stage with Peltier elements. The EQE is performed using the PV-tools LOANA that includes a temperature-controlled stage.

Encapsulation

Solar ribbons are connected to the contact of the tandem with silver paste to extend the electrodes. The device is laminated between 2 layers of encapsulant (TPU) and 2 glasses. Butyl rubber serves as an edge sealant to avoid moisture ingress from the sides. The lamination (Ecolam5, Ecoprogetti) temperature is set at 120°C for 25mn.

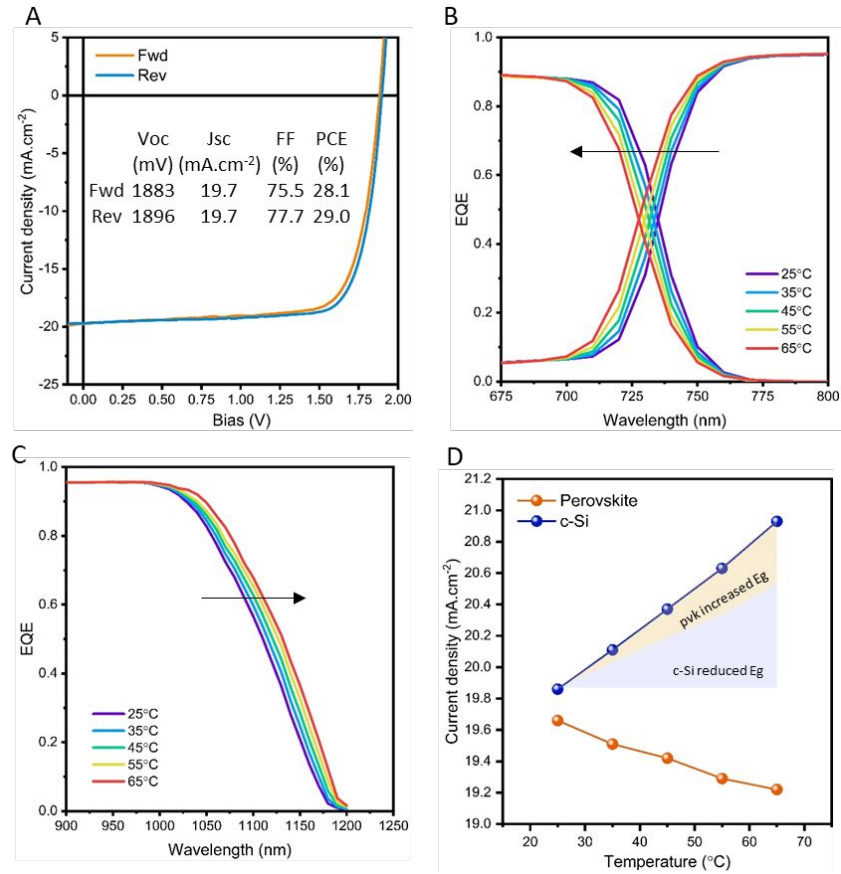


Figure S1. (A) Device performance under STC (25°C, AM1.5G and 100 mW.cm⁻²) (B) Detailed EQEs at the perovskite edge absorption at different temperatures (C) Detailed EQEs at the silicon edge absorption at different temperatures (D) Current densities extracted from the EQEs at different temperature based on AM1.5G spectrum.

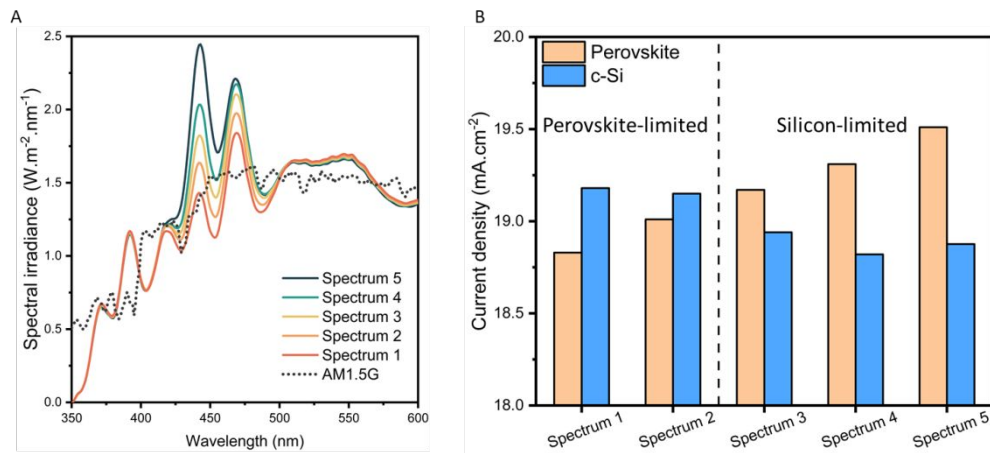


Figure S2. (A) Detailed spectra around 450 nm generated for this study. (B) Current densities expected at 25°C from the 5 spectra. The tandem is perovskite-limited for spectra 1 and 2. The tandem is silicon-limited for spectra 3,4 and 5.

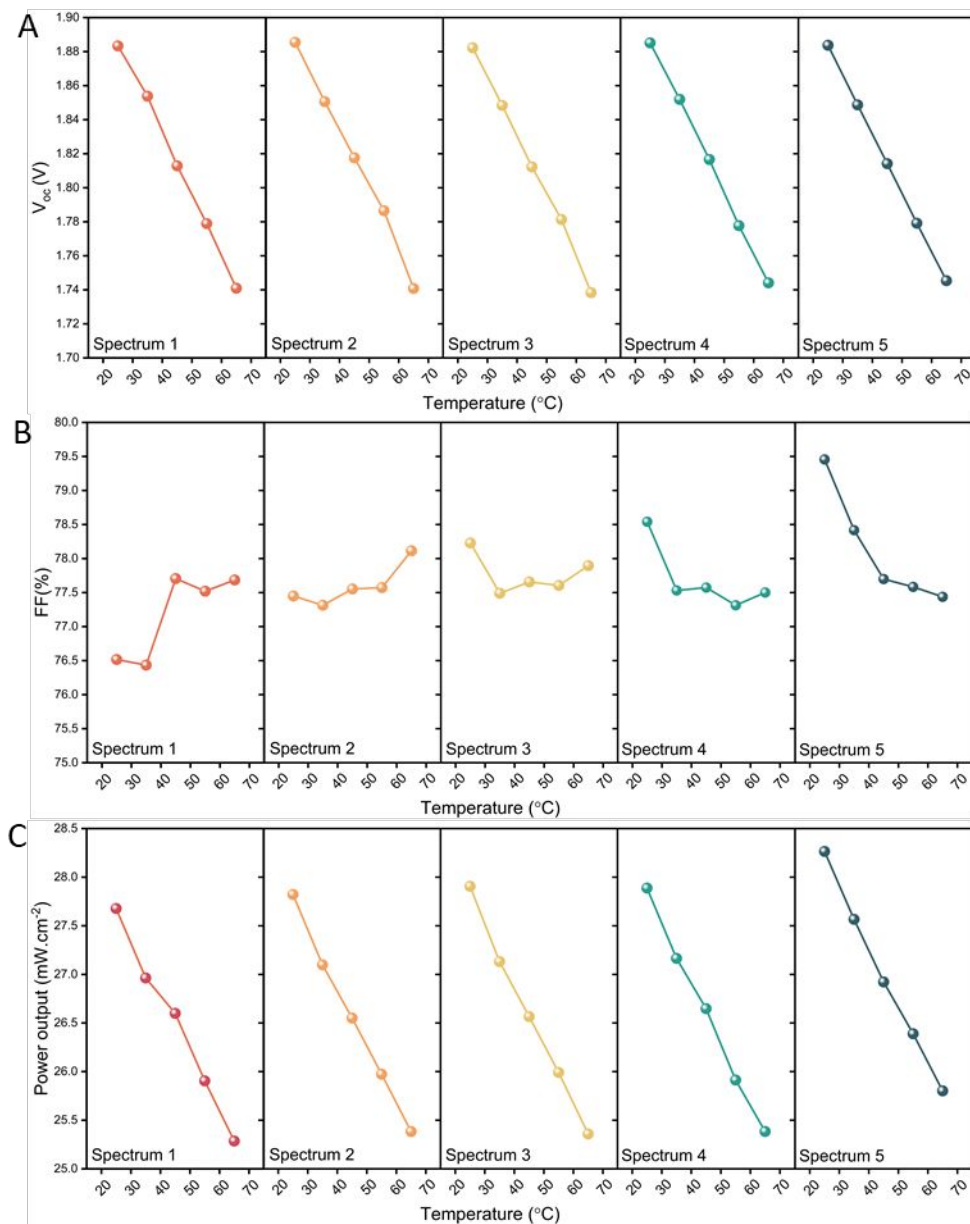


Figure S3. (A) V_{oc} (B) FF and (C) PCE of the tandem depending on the spectrum and the temperature.

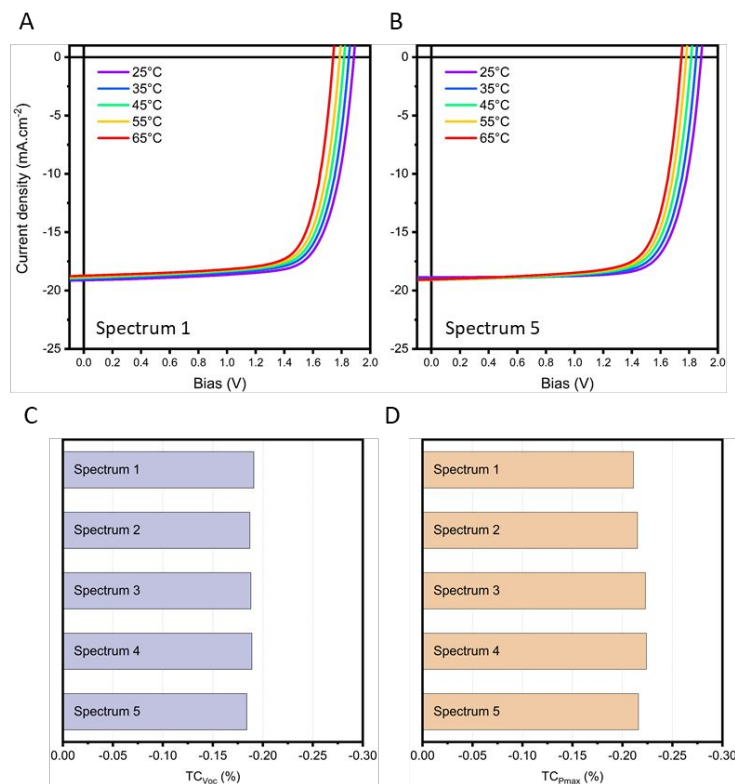


Figure S4. JV scans (reverse) of the tandem at different temperatures for (A) spectrum 1, perovskite-limited condition and (B) spectrum 5, silicon-limited condition. (C) Open circuit temperature coefficient of the tandem under different spectra. (D) P_{max} temperature coefficient of the tandem under different spectra.

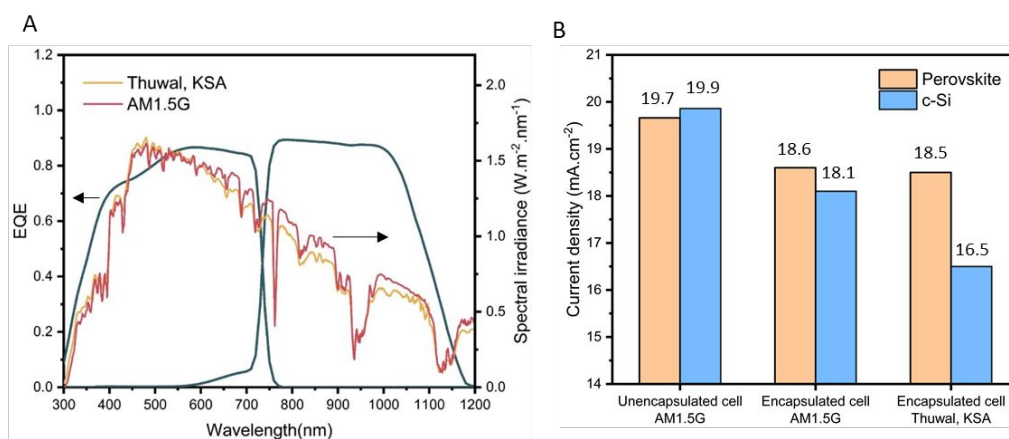


Figure S5. (A) EQE of the encapsulated tandem used for the field measurement as well as the AM1.5G spectrum and local spectrum measured on-site in KAUST location. (B) Expected subcell current densities at 25°C from the EQEs before and after encapsulation based on AM1.5G and local spectrum.